

12. Research Case Studies

*“Without data you're just
another person with an opinion.”
— W. Edwards Deming*

Having learned data analytics through the data science process, this chapter presents case studies from the author's own research on data analytics, published in international conferences and journals.

12.1 Detecting Road Damage with a Gyro Sensor

Drawn from “Monitoring Roadway Lights and Pavement Defects for Nighttime Street Safety Assessment by Sensor Data Analysis and Visualization,” published in *Sensors and Materials*, Vol. 30, No. 10 (2018), special issue on *Sensors and Materials for Cyber-Physical Applications*, MYU K.K., Japan (Chawuthai, 2018).

Research Problem

This research aimed to build a model detecting road damage using a gyro sensor mounted on a moving vehicle, distinguishing potholes and bumps in the road surface from speed bumps — a classification analysis problem.

Data

An IoT device was built with a Raspberry Pi connected to a gyro sensor and an accelerometer. The gyro sensor measured rotation on the x, y, and z axes (gX, gY, gZ), and the accelerometer measured acceleration on 3 axes (aX, aY, aZ). Driving around to collect data, sensor readings — along with timestamp (ts) and GPS coordinates (lat, lon) — were sent to a Firebase cloud database every 500 milliseconds. Each point was labeled as normal road or speed bump (defect=0) or damaged road (defect=1), as in Table 12.1-1, totaling 20,000 rows.

Table 12.1-1: Example sensor data for road-damage detection

ts	lat	lon	aX	aY	aZ	gX	gY	gZ	defect
2017-09-20 22:22:06.5	14.64088	100.20991	-0.364	0.466	0.985	-8.940	2.720	18.81	0
2017-09-20 22:22:07.0	14.64081	100.20975	-1.963	-0.956	6.447	-2.500	-6.110	21.99	0
2017-09-20 22:22:07.5	14.64072	100.20958	0.623	-0.285	-7.581	18.14	-19.92	4.360	1
2017-09-20 22:22:08.0	14.64063	100.20940	1.004	1.510	-3.330	-1.710	-4.760	5.310	0

Data Analysis

This classification problem uses aX, aY, aZ, gX, gY, and gZ as features to predict defect, testing:

- Logistic Regression
- Decision Tree, using the GINI Coefficient as the split-quality measure
- Artificial Neural Network, with 1 hidden layer of 10 nodes, logistic activation, max 5,000 training iterations, stopping tolerance 0.0001, and learning rate 0.001

Analysis used all sensors together, then the gyro and accelerometer sensors separately, evaluated via 4-Fold Cross-Validation with F1 (F-Measure).

Results

Table 12.1-2 shows Decision Tree gave the best-performing model, using only the gyro sensor — meaning road-damage detection can rely solely on the gyro sensor's x, y, z rotation data. The resulting model was also tested in the field, successfully mapping detected damage locations (Figure 12.1-1).

Table 12.1-2: F1 results for road-damage detection

Technique	aX,aY,aZ,gX,gY,gZ	aX,aY,aZ	gX,gY,gZ
Logistic Regression	0.716	0.527	0.741
Decision Tree	0.821	0.630	0.918 (best)
Artificial Neural Network	0.763	0.452	0.872

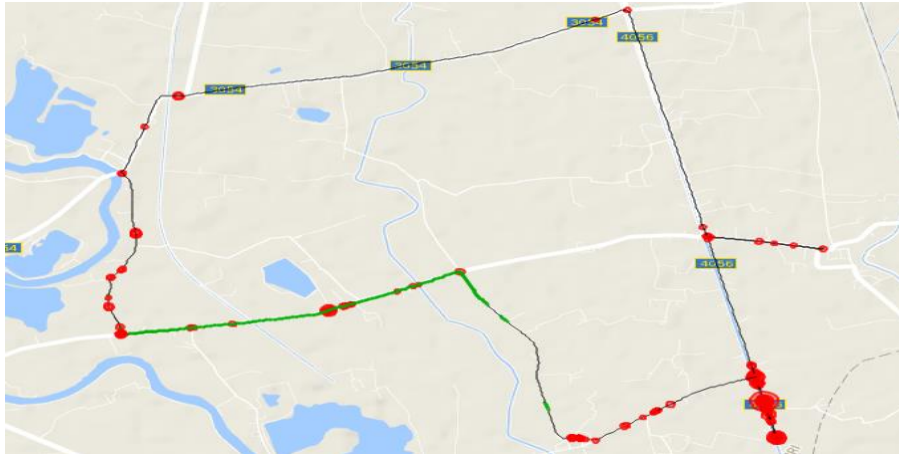


Figure 12.1-1: Road-damage locations detected by the developed Decision Tree model

12.2 Detecting Kicks with a Microwave Sensor

Drawn from “The Analysis of a Microwave Sensor Signal for Detecting a Kick Gesture,” presented at the International Conference on Engineering, Applied Sciences, and Technology (ICEAST), published by IEEE, 2018 (Chawuthai & Sakdanuphab, 2018).

Research Problem

This research built a device to detect a kicking gesture, to open a car's trunk hands-free when the user's hands are full. It used a microwave sensor emitting waves that reflect off objects back to a receiver, computing object speed via the Doppler Effect. Volunteers' activities — walking by, kicking, or other objects passing — were recorded and their signals examined, then processed for classification analysis with two outcomes: kick or not kick.

Data

Signals were recorded across 3 activity types: kicking, walking, and something passing. Exploring the data revealed clearly distinguishable signal shapes (Figure 12.2-1), suitable for classification.

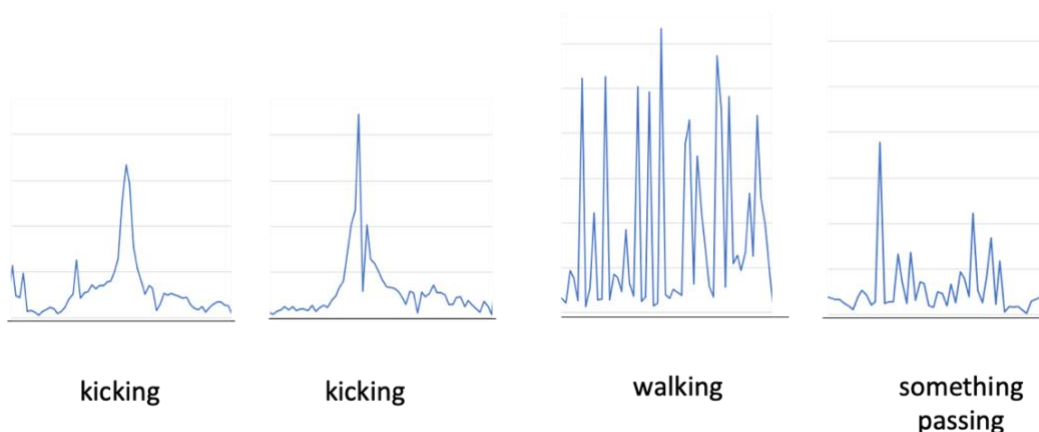


Figure 12.2-1: Microwave sensor signals across different activities

Table 12.2-1: Example microwave signal data for kick detection

t001	t002	t003	t004	t005	t006	t007	...	t100	kick
20	20	20	20	20	20	20	...	20	0
20	20	20	20	20	20	15	...	15	0
20	20	20	20	15	11	8	...	15	0
20	20	20	15	11	8	7	...	20	1
20	20	15	11	8	7	7	...	20	1
20	15	11	8	7	7	8	...	20	1

From the raw signal stream (10ms per reading), a sliding window of 1,000ms (100 readings) formed each set of input features, with output labeled kick (kick=1) or not (kick=0, covering idle or other activities like walking or objects passing), as in Table 12.2-1.

Data Analysis

Classification techniques tested, evaluated via 4-Fold Cross-Validation with Accuracy:

- Logistic Regression
- Decision Tree, using the GINI Coefficient
- Artificial Neural Network, 1 hidden layer of 10 nodes, logistic activation, max 5,000 iterations, stopping tolerance 0.0001, learning rate 0.001
- Support Vector Machine, RBF (Radial Basis Function) kernel, gamma 0.3, max 5,000 iterations, stopping tolerance 0.0001

Results

Logistic Regression gave the highest accuracy, 0.984 (Table 12.2-2). The final model was then trained on the full dataset, with its 100 feature coefficients and intercept implemented in C on a microcontroller for real-world deployment.

Table 12.2-2: Kick-detection evaluation results

Technique	Accuracy
Logistic Regression	0.984 (best)
Decision Tree	0.822
Artificial Neural Network	0.941
Support Vector Machine	0.887

12.3 A Recommender for Biologists Studying Fungi–Host Pairs

Drawn from “Link Prediction in Linked Data of Interspecies Interactions Using Hybrid Recommendation Approach,” presented at the Joint International Semantic Technology Conference, published in Lecture Notes in Computer Science (LNCS), Springer, 2014 (Chawuthai et al., 2014; Chawuthai, 2016).

Research Problem

This research modeled relationships between observed fungi and their hosts (plants or animals), to recommend to mycology researchers where else a given fungus might be found on other hosts — framed as a prescriptive analytics problem.

Data

Original data was scanned from the book *A List of Fungi Recorded in Japan*: 9,151 fungi–host pairs, covering 3,884 fungus species and 2,582 host species. The book listed both scientific and Japanese common names (e.g., モミ, scientific name *Abies firma*), which were converted entirely to scientific names, as in Table 12.3-1.

Table 12.3-1: Example fungi–host observation data

fungi	host
<i>Blastospora itoana</i>	<i>Prunus grayana</i>
<i>Blastospora itoana</i>	<i>Prunus persica</i>
<i>Puccinia coronata</i>	<i>Triticum aestivum</i>
<i>Puccinia acetosae</i>	<i>Rumex acetosella</i>
<i>Uredinopsis filicina</i>	<i>Abies firma</i>

Data Analysis

Analysis used link prediction, viewing the fungi–host relationship as a bipartite graph, and built models to predict the probability of missing links using 3 approaches:

- Collaborative Filtering, recommending likely relationships, named model P^{cf} (also used as the baseline for comparison).
- Community Detection, grouping fungi and hosts into social communities based on similar relationship patterns, then estimating the probability of a relationship between a fungus and another host in the same community — named model P^{cs} .
- Content-Based Filtering, recommending likely relationships based on genus-level similarity between fungus and host species — named model P^{Bc} .

These 3 models were then combined into a Hybrid Model via linear learning, named P^{ii} :

$$P^{ii}(f,h) = 0.763 \times P^{cf}(f,h) + 0.543 \times P^{cs}(f,h) + 0.101 \times P^{Bc}(f,h) \\ + 0.428 \times P^{cf}(h,f) + 0.584 \times P^{cs}(h,f) + 0.025 \times P^{Bc}(h,f)$$

Results

Testing via Train-Test Split, with 20% of data randomly held out for testing, P^{ii} outperformed the baseline model, evaluated via Top-100 Precision alongside AUC (Area Under the receiver operating characteristic Curve), as in Table 12.3-2.

The highest-scoring, not-yet-observed fungi–host relationships recommended by P^{ii} were further investigated, and several were confirmed to genuinely exist, discovered by researchers at Japan's National Museum of Nature and Science after being guided by this research's recommendations, including:

- Fungus *Amanita pseudoporphyria* found on host *Megaselia salteri*
- Fungus *Russula alboareolata* found on host *Drosophila brachynephros*
- Fungus *Amanita kotohiraensis* found on host *Megaselia gotoi*
- Fungus *Tylophilus ballouii* found on host *Megaselia flava*
- Fungus *Tylophilus ballouii* found on host *Tricimba japonica*

Table 12.3-2: Test results for the fungi–host recommendation models

Recommendation Technique	Top-100 Precision	AUC
$P^{cf}(f,h)$ — baseline model	0.303	0.855
$P^{ii}(f,h) = 0.763 \times P^{cf}(f,h) + 0.543 \times P^{cs}(f,h) + 0.101 \times P^{Bc}(f,h) + 0.428 \times P^{cf}(h,f) + 0.584 \times P^{cs}(h,f) + 0.025 \times P^{Bc}(h,f)$ — best	0.577 (best)	0.906 (best)

12.4 A Recommender for Vehicle Rest Stops

Drawn from “A Hybrid Method for Predicting a Potential Next Rest Stop of Commercial Vehicles,” presented at the International Symposia of Transport Simulation (ISTS) and the International Workshop on Traffic Data Collection and its Standardization (IWTDCS), Japan, published in Transportation Research Procedia, Elsevier, 2018 (Chawuthai et al., 2018).

Research Problem

This research predicted commercial vehicles' next rest stop, helping anticipate where vehicles will stop next to prepare service and readiness — framed as a prescriptive analytics problem.

Data

Data came from commercial vehicle GPS tracking — coordinates (lat, lon) and timestamp (ts) — along with vehicle type, as in Table 12.4-1.

Table 12.4-1: Example vehicle GPS tracking data

vid	ts	lat	lon	vtype
0001	2018-06-06 01:30:00	9.638598	99.12456	Coach
0002	2018-06-06 01:30:00	7.867501	98.35564	Coach
0003	2018-06-06 01:30:00	7.903763	98.30505	Bus
0004	2018-06-06 01:30:00	8.558653	99.34329	Truck
0005	2018-06-06 01:30:00	8.120136	98.33496	Coach
0006	2018-06-06 01:30:00	9.6489	99.12369	Bus
0007	2018-06-06 01:30:00	9.638598	99.12456	Truck

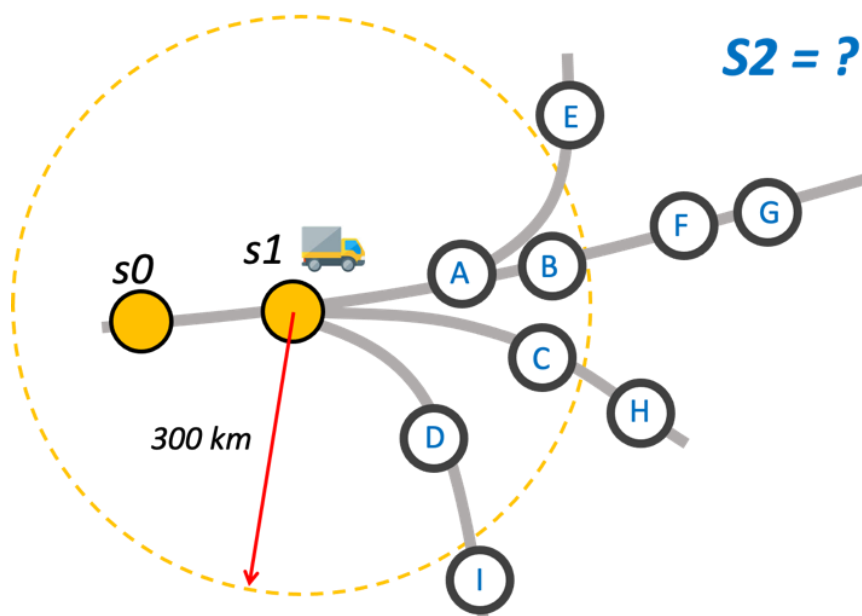


Figure 12.4-1: Diagram of next-stop selection

Data Analysis

The approach used the previous stop (s_0) and current stop (s_1) to predict the next stop (s_2), where a stop was defined as a 30-minute to 3-hour stay after 50–300 km of travel, at a location visited by at least 100 vehicles per day. Given a current stop (s_1), the model considered all candidate stops within 300 km (Figure 12.4-1), which shows the previous stop (s_0), current stop (s_1), and candidate next stops A, B, C, D.

Table 12.4-2: Example stop-sequence data for analysis

s0	s1	s2	vtype	ts1
(13.73, 100.76)	(13.12, 100.91)	(13.73, 100.76)	Bus	2017-10-06 10:37
(13.05, 100.89)	(13.73, 100.76)	(13.05, 100.89)	Bus	2017-10-11 12:47
(13.73, 100.76)	(13.10, 100.90)	(13.73, 100.76)	Coach	2017-09-28 03:02
(13.73, 100.76)	(13.05, 100.89)	(13.74, 100.76)	Coach	2017-10-07 09:18
(13.10, 100.90)	(13.12, 100.91)	(13.10, 100.90)	Coach	2017-09-20 19:02
(13.10, 100.90)	(13.05, 100.90)	(13.10, 100.90)	Coach	2017-09-25 21:28
(13.73, 100.76)	(13.75, 100.76)	(13.73, 100.76)	Bus	2017-09-28 10:24
(13.71, 100.76)	(13.65, 100.78)	(13.71, 100.76)	Truck	2017-09-22 14:35
(13.12, 100.91)	(13.73, 100.76)	(13.05, 100.89)	Coach	2017-09-28 09:15

Raw GPS data (Table 12.4-1) was processed to identify each vehicle's stops and their sequence, recorded as in Table 12.4-2, with s0, s1, s2 as stop coordinates (s2 the target to predict), vtype the vehicle type, and ts1 the timestamp at the current stop (s1).

This data then fed a Hybrid Recommender System combining 3 scoring models:

- Model SF — estimates the probability of stopping at s2 given a history of stopping at s0 then s1.
- Model SD — considers the vector directions from s0 to s1 and s1 to s2 using Cosine Similarity: a straight-line path across all three points scores higher than an angled one, and a sharp turn (resembling backtracking) scores very low.
- Model SP — counts each stop's overall popularity, weighted against the popularity of all stops.

These models were learned and combined into a linear hybrid recommender:

$$P^{Next}(s0, s1, s2^*) = SF(s0, s1, s2^*) + 0.8SD(s0, s1, s2^*) + 0.2SP(s0, s1, s2^*)$$

where P^{Next} scores a candidate next stop $s2^*$, given the known previous stop (s0) and current stop (s1).

Results

Tested via Train-Test Split and evaluated with AUC, the model achieved 0.848 — a good-to-very-good result for a recommender system.

Breaking results down by vehicle type and time of day: liquid-cargo trucks driving during the day scored highest (AUC 0.938), followed by scheduled buses (day and night), container trucks (day and night), and hazardous-cargo trucks driving during the day.

12.5 Chapter Summary

This chapter presented 4 of the author's own data-analytics research projects. The first two — “Detecting Road Damage with a Gyro Sensor” and “Detecting Kicks with a Microwave Sensor” — are predictive-analytics examples in classification, with sensor data as features and nominal-scale group labels as the target. The latter two — “A Recommender for Biologists Studying Fungi–Host Pairs” and “A Recommender for Vehicle Rest Stops” — are prescriptive-analytics examples, blending classification and clustering science into working hybrid recommender systems. All these projects illustrate applying data analytics to real problems: framing the problem clearly, choosing a trustworthy evaluation method, understanding and preparing the data, selecting suitable analytics techniques, and reporting model results.

12.6 Review Questions

1. From “Detecting Road Damage with a Gyro Sensor,” discuss what type of problem, what data, and what evaluation method would suit extending this research to measure pothole severity.
2. From “Detecting Kicks with a Microwave Sensor,” discuss what would happen if another activity later produced a similar signal to kicking, and how the device could be improved.
3. From “A Recommender for Biologists Studying Fungi–Host Pairs,” discuss why using Collaborative Filtering alone performed worse than combining it with other techniques.
4. From “A Recommender for Vehicle Rest Stops,” discuss what additional variables could improve this recommender's performance.
5. From “A Recommender for Vehicle Rest Stops,” discuss how evaluating results separately by vehicle type benefits the analytics work.

12.7 References

Chawuthai, R. (2016). Linked Open Data-Based Knowledge Management Process for Biodiversity. Doctoral dissertation, SOKENDAI.

Chawuthai, R. (2018). Monitoring Roadway Lights and Pavement Defects for Nighttime Street Safety Assessment by Sensor Data Analysis and Visualization. *Sens. Mater*, 30, 2267–2279.

Chawuthai, R., & Sakdanuphab, R. (2018). The Analysis of a Microwave Sensor Signal for Detecting a Kick Gesture. In 2018 International Conference on Engineering, Applied Sciences, and Technology (ICEAST) (pp. 1–4). IEEE.

Chawuthai, R., Chankaew, N., & Threepak, T. (2018). A Hybrid Method for Predicting a Potential Next Rest Stop of Commercial Vehicles. *Transportation Research Procedia*, 34, 36–43.

Chawuthai, R., Takeda, H., & Hosoya, T. (2014). Link Prediction in Linked Data of Interspecies Interactions Using Hybrid Recommendation Approach. In Joint International Semantic Technology Conference (pp. 113–128). Springer.

Lü, L., & Zhou, T. (2011). Link Prediction in Complex Networks: A Survey. *Physica A: Statistical Mechanics and Its Applications*, 390(6), 1150–1170.